

DFD Symbol Legend

Symbol	Meaning	Used in this Project
Rectangle	External Entity	Farmer
Rounded Rectangle	Process	Smart Agricultural Engine, Crop Recommendation
Open Rectangle	Data Store	Soil Dataset, Weather Dataset, ML Crop Prediction Model
Arrow	Data Flow	Soil & Weather Data, Recommended Crop & Insights

Data Flow Process

1. Farmer

- Opens the application.
- Enters soil and weather details.

2. Smart Agricultural Engine

- Receives the input data.
- Validates and processes the information.

3. Data Stores

- Soil Data Dataset
- Weather Data Dataset
- Machine Learning Crop Prediction Model

4. Crop Recommendation

- AI analyzes the input using the trained ML model.
- Generates the most suitable crop recommendation.

5. Output

- Displays the recommended crop.
- Provides farming insights and prediction confidence.

Summary

The DFD illustrates the flow of data from the **Farmer** to the **Smart Agricultural Engine**, which uses the **Soil Dataset**, **Weather Dataset**, and **ML Model** to generate an accurate **Crop Recommendation** and useful farming insights.

Data Flow Diagram

